

Variables and console input

CSC103 Programming Fundamentals / Lecture 04

Dr. Muhammad Farhan

Associate Professor of Computer Science
Department of Computer Science
COMSATS University Islamabad
Sahiwal Campus

Fall 2026

The problem for this lecture

Read a quantity and price using Scanner. Print their product. Then read a one-letter category. Use quantity 3, price 12.5, category A.

Create one Scanner and read quantity, price and category in order.

Learning objectives

- Declare and initialise variables and constants
- Read numeric and character input
- Trace assignment as a change of stored value

CLO-1

Variables and initialisation

Assignment copies a value

Scanner input

Reading one character

Variables and initialisation

- A variable has a declared type and a name.
- Initialisation supplies its first value. A local variable must receive a value before a read.

Example

```
int seats = 30;  
seats = 28;  
final int ROOM_LIMIT = 40;
```

Assignment copies a value

- The right-hand expression is evaluated before assignment.
- For primitive variables, assigning one variable to another copies the current primitive value.

Example

```
int a = 7;  
int b = a;  
a = 9;  
// b is still 7
```

Scanner input

- Create one Scanner for console input.
- `nextInt` reads an integer token and `nextDouble` reads a decimal token.
- The entered token must match the requested type.

Example

```
java.util.Scanner in =  
    new java.util.Scanner(System.in);  
int age = in.nextInt();
```

Reading one character

- Scanner has no nextChar method.
- next().charAt(0) reads the first UTF-16 code unit of the next token.
- For this exercise, input is one basic Latin character.

Example

```
String token = in.next();  
char choice = token.charAt(0);
```

Choosing a type from the data

- A quantity that counts whole items fits an integer type.
- A price with a fractional part needs a suitable numeric representation.
- For introductory arithmetic, double is convenient, but exact financial applications need decimal handling.

Example

quantity: 3

price: 12.5

category: A

Each input has a different role and type.

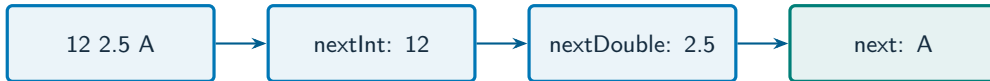
Console input is a sequence of tokens

- Scanner reads the next available token when a next method succeeds.
- The program must request the token type in the order the user supplies it.
- A character read through `next().charAt(0)` uses only the first character of that token.

Example

```
Input tokens: 3 12.5 A
nextInt consumes 3
nextDouble consumes 12.5
next consumes A
```

Input tokens become typed values



What to notice

Each Scanner call consumes the next token according to the requested type.

Key distinctions

Declaration	Initial value	May be reassigned?
int quantity = 3	3	Yes
double price = 12.5	12.5	Yes
char category = 'A'	A	Yes
final int LIMIT = 20	20	No after initial assignment

These distinctions determine which operation or control structure fits the problem.

First example: execution

```
int initial = 10;  
int remaining = initial;  
remaining = remaining - 3;  
final int CAPACITY = 20;  
System.out.println(remaining);  
System.out.println(initial);
```

First example: result

7

10

Assigning remaining does not change initial.

Scanner input

Reading one character

Guided practice

Read a quantity and price using Scanner. Print their product. Then read a one-letter category. Use quantity 3, price 12.5, category A.

Attempt the solution before continuing.
Use the definitions and examples from this lecture.

Solution strategy

- Create one Scanner and read quantity, price and category in order.
- Calculate quantity * price using the numeric values.
- Print the product and the selected category.

Sample console input
3 12.5 A

Complete solution

```
public class Solution04 {  
    public static void main(String[] args) throws Exception {  
        java.util.Scanner in = new java.util.Scanner(System.in);  
        in.useLocale(java.util.Locale.US);  
        int quantity = in.nextInt();  
        double price = in.nextDouble();  
        char category = in.next().charAt(0);  
        double total = quantity * price;  
        System.out.println(total);  
        System.out.println(category);  
    }  
}
```


Execution trace

Read or calculation	Consumed token	Stored result
nextInt()	3	quantity = 3
nextDouble()	12.5	price = 12.5
next().charAt(0)	A	category = 'A'
quantity * price	No new input	total = 37.5

Follow the changing state in execution order.

Solution result

37.5
A

Why this result follows
Print the product and the selected
category.

A common incorrect approach

```
int total;  
System.out.println(total);
```

Example

A local variable has no usable automatic initial value. Assign total before printing it, **for**
example **int** total = 0.

Boundary and failure checks

Check the stated input assumptions.
Read a rectangle length and width. Print area and perimeter. State the units and allowed input range.

Use `int quantity = in.nextInt()`,
`double price = in.nextDouble()`, and
`char category = in.next().charAt(0)`.
Product: 37.5. Category: A.

Check your understanding

After `int x=4; int y=x; y=8;`, what is `x`? Can a final variable be assigned again?

Write a prediction and explain the rule that supports it.

Answer and explanation

x remains 4. A final variable cannot be reassigned after its initial assignment.

A local variable has no usable automatic initial value. Assign total before printing it, for example `int total = 0`.

Compare cases and predict outcomes

Next token	Requested operation	Result
12	<code>nextInt()</code>	int value 12
2.5	<code>nextDouble()</code>	double value 2.5
A	<code>next().charAt(0)</code>	char value A

Discuss

Explain each expected result before running the example. Which case would reveal a wrong assumption?

Apply the idea

Independent practice

A shop sells 12 items at 2.5 each. Show the variable declarations and total. Then predict what happens if `nextInt()` is used for the token 2.5.

1. Work through the input by hand.
2. Write or correct the program.
3. Justify the expected result.

Solution and reasoning

```
int quantity = 12;  
double unitPrice = 2.5;  
double total = quantity * unitPrice;  
System.out.println(total);
```

- The int operand is promoted for multiplication with double.
- The result is 30.0.
- Scanner.nextInt() cannot consume 2.5 as an int and throws InputMismatchException.

Circle area with console input

- Declaration introduces radius and area; assignment gives a local variable a usable value.
- A named constant documents the chosen approximation to pi.
- Read the radius as double, then calculate only after input is available.

Source: [supplied ISB campus slides](#). See [speaker notes](#) and [ISB_Source_Map.md](#).

Worked problem: Circle area with console input

Task

Read radius 20 and calculate its area with $PI = 3.14159$, matching the source example. Assume a nonnegative radius.

Input: 20

Before running

Predict the output and identify the variables that change.

Complete ISB example (1/1)

```
public class IsbLecture04 {  
    public static void main(String[] args) throws Exception {  
        java.util.Scanner input = new java.util.Scanner(System.in);  
        input.useLocale(java.util.Locale.US);  
        final double PI = 3.14159;  
        double radius = input.nextDouble();  
        double area = PI * radius * radius;  
        System.out.printf(java.util.Locale.US, "Area: %.3f%n", area);  
    }  
}
```

Worked trace and expected result

Moment	radius	area
After reading	20.0	Not assigned yet
After calculation	20.0	1256.636
Radius changed to zero	0.0	Recalculation gives 0.000

Expected console output

Area: 1256.636

Extend the ISB example

Practice

Why does changing radius after calculating area not automatically change area?

Explain your reasoning

Assignment stores the computed value at that moment. Recompute area after changing radius.

Lecture summary

- and initialise variables and constants
 - numeric and character input
 - assignment as a change of stored value
- Create one Scanner and read quantity, price and category in order.
 - Calculate quantity * price using the numeric values.
 - Print the product and the selected category.

After-class practice

Read a rectangle length and width. Print area and perimeter. State the units and allowed input range.

Syllabus reading

Liang Ch 2

Next lecture: Types and arithmetic expressions